Multi-Omics Target Discovery Package: Alzheimer's Disease

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Repository: github.com/sharathmandula45-cmyk/biot6900 (Labs 2–4 notebooks, targets_ad.csv, targets_ad_w3.csv, structure_assessment_ad.csv, figures)

1. Disease and question

Alzheimer's disease (AD) is the most common cause of dementia. After many years of research, only a few treatments that can slow down disease progression have reached patients, and most of them target amyloid. Therefore, researchers need to find new drug targets beyond amyloid using reliable human data. In this project, a good AD target must pass three tests, one for each lab: (1) it must be supported by more than one independent type of human AD evidence, such as RNA, protein, and genetic data (Lab 2); (2) it must have good potential for drug development, including the suitable drug type or modality (Lab 3); and (3) its predicted structure must be reliable enough to support a claim about a possible drug-binding site (Lab 4). After applying these criteria, the analysis ends with five shortlisted genes, with SAMHD1 selected as the main lead target.

2. Multi-omics integration: from ~22,000 genes to 1,139

I combined three AD datasets that did not use the same patients: hippocampal RNA data (GSE5281; Liang et al., 2007), cortical TMT protein data (Johnson et al., 2022), and genetic data from the GWAS Catalog (Sollis et al., 2023). The datasets were matched using gene symbols after separating GWAS entries that contained multiple genes. The Venn diagram (Figure 1A) shows how the number of genes decreased during this process. The RNA dataset covered 22,189 genes, but only 7,858 proteins had usable fold-change and p-value information, while 4,108 genes had a mapped GWAS hit. Only 1,139 genes were present in all three evidence layers. The GWAS data acted as the strictest filter because 6,287 genes had both RNA and protein evidence but no GWAS signal. The diagram also shows how some known AD-related genes were excluded before scoring. For example, TREM2 and CD33 were present in the RNA and GWAS datasets but fell into the 1,308-gene RNA-plus-GWAS-only group because their protein data did not have usable fold-change values.

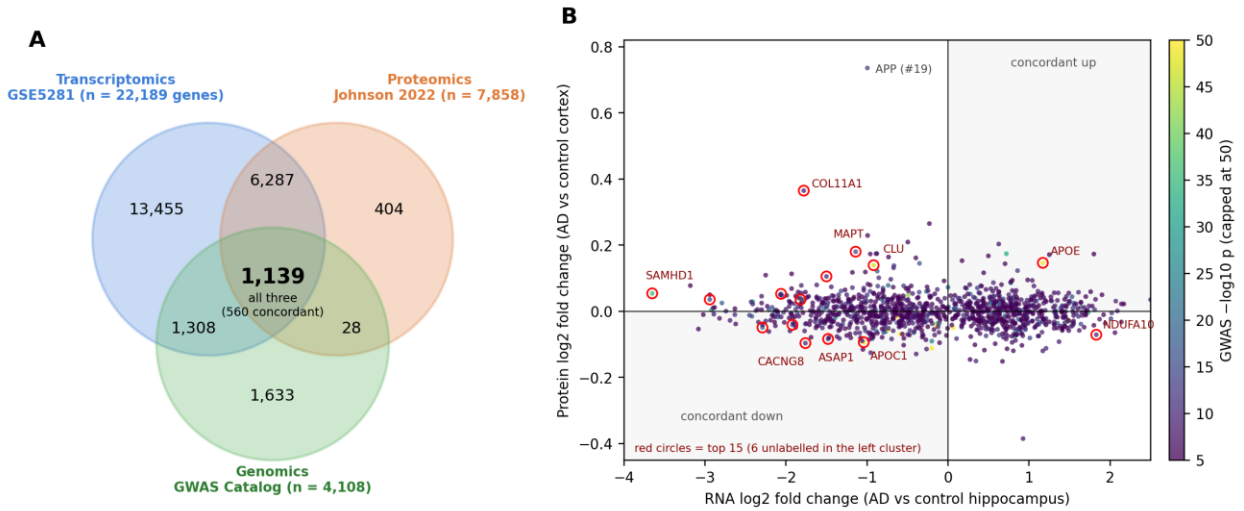


Figure 1. Multi-omics integration. *(A) Venn diagram of genes with usable evidence in each layer before the join (rows with a gene symbol and p-value). The centre is the 1,139-gene three-way join used for scoring. (B) Integration of all three layers for the 1,139 genes: RNA vs protein log2 fold change, coloured by GWAS strength. Shaded quadrants are sign-concordant. Red circles mark the top 15 by multi_evidence_score. Regenerated from the Lab 2 merged table (plotting only, no new analysis).*

Concordance. Only 560 out of 1,139 genes, or 49%, showed changes in the same direction at both RNA and protein levels. This is almost the same as the 50% expected by chance. Figure 1B helps explain this result. RNA fold changes ranged from approximately −3.7 to +2.2, whereas most protein fold changes were within ±0.2. The RNA data came from 23 hippocampal donors, while the protein data came from a different and much larger group of people and were collected from the prefrontal cortex. Therefore, the two datasets do not measure exactly the same biological conditions. For this reason, I used agreement between RNA and protein results as something to explain and investigate rather than as a strict filtering rule.

Known-gene recovery: The ranking also identified several well-known AD-related genes, even though they were not specifically selected in advance. APOE ranked fourth, MAPT, which is related to tau, ranked eighth, APOC1 ranked ninth, and CLU ranked

thirteenth. APP ranked nineteenth and showed the largest protein increase among the highly ranked genes, with a log2 fold change of +0.74. Other established AD-related genes, including BIN1 at rank 61, PLCG2 at rank 154, and PICALM at rank 221, were also retained after joining the datasets but received lower rankings because their protein changes were close to zero. This suggests that the analysis can identify known AD-related signals while also helping researchers find less obvious potential targets. These findings make the newly identified candidates worth further investigation, while additional experiments are still needed to confirm their roles in Alzheimer's disease.

3. Druggability triage: which candidates look druggable, and how

The top 15 genes were analysed using the Lab 3 protein language model to estimate their drug-development potential. ESM C (300M) embeddings (ESM Team, 2024) were used as input for a logistic regression model trained on 1,201 proteins with clinical drug-development labels from Open Targets (Ochoa et al., 2023). The model gave two scores: small-molecule potential (p_sm) and antibody potential (p_ab). One gene, CACNG8, was already present in the training dataset with a clinical label for small-molecule drugs. To avoid data leakage, CACNG8 was removed from the training data, leaving 1,200 proteins for model training. The most reliable performance result came from the family-based split, which gave an AUROC of 0.728 and an average precision (AP) of 0.531, compared with a chance AP of 0.333. This AUROC was about 0.10 lower than the random-split result, showing that the model is useful but only moderately accurate when predicting druggability for new protein families. Therefore, disease-related evidence was given more importance, with a weight of 0.6. The model results showed only weak agreement with the Lab 2 gene ranking (Spearman correlation ≈ 0.14). This is understandable because the two methods answer different questions. Lab 2 identifies genes associated with AD, while Lab 3 estimates whether a protein is similar to proteins that have already been targeted by drugs.

Figure 2 shows how each candidate performed in both labs. SAMHD1 was the only target with a strong small-molecule score (p_sm = 0.824). Its nearest drugged protein neighbours were PDE1B, CPT1B, and PARP1. This result fits SAMHD1 because it is an intracellular enzyme with nucleotide-binding sites that may provide opportunities for small-molecule drug development. CACNG8 (p_sm = 0.380, with its own clinical label hidden during prediction) and, to a lesser extent, ASAP1 (p_sm = 0.193) also remained possible small-molecule targets. APOC1 (p_ab = 0.248) and CLU (p_ab = 0.131) showed a greater preference for antibody-based approaches, which is reasonable because both proteins are secreted. However, MAPT and APOE received very low scores for both drug types. MAPT, which encodes the disordered tau protein, dropped from rank 8 to rank 14, even though anti-tau antibodies have already reached clinical trials. APOE also scored very low, with p_sm = 0.002 and p_ab = 0.016, partly because very few apolipoproteins have drugs that have reached the clinical stage. These results show that a low model score does not necessarily mean a protein cannot be targeted. It may simply mean that few similar proteins have been successfully developed as drugs. Based on these findings, SAMHD1 was selected as the strongest candidate, while APOE was chosen to test a possible weakness of the model. Both genes were then taken forward for structural analysis in Lab 4.

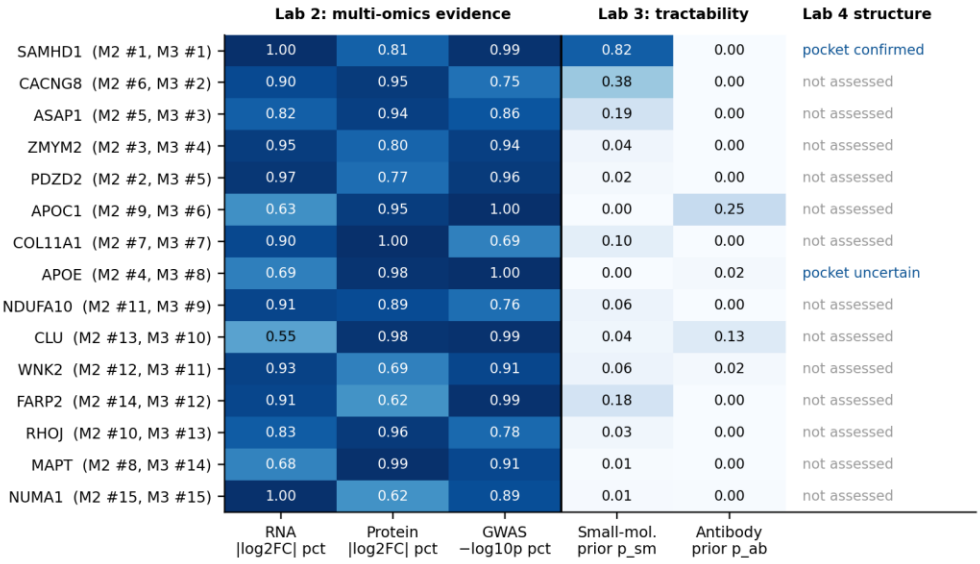


Figure 2. Target prioritisation across all three labs. *Rows are the top 15 sorted by the Lab 3 final rank (w = 0.6). Lab 2 columns are rank percentiles among the 1,139 joined genes; Lab 3 columns are the ESM C tractability priors; the right column is the Lab 4 structure call. M2/M3 = Module 2/3 rank.*

4. Structural assessment: is there a pocket I can trust?

Target	Mean pLDDT (% <50)	Domains / inter-domain PAE	RMSD vs experiment	Best kept pocket (fpocket)	Call
SAMHD1 Q9Y3Z3	88.2 (9.6%); HD domain ~115–600 mostly >90	1 domain (38–600); SAM domain placement less certain	**0.67 Å**, 472 Cα pairs vs 4BZC (X-ray, dGTP-bound)	Pocket 3: score 0.483, 1,227 Å^3, pLDDT 96.6, PAE 1.9 Å; next best 0.026	**Structure supports a small-molecule pocket**
APOE P02649	75.5 (16.1%); 1–40 disordered	2 domains (42–208, 226–303); **25.8 Å** between them	32.35 Å overall (meaningless); **2.60 Å** on the 4-helix bundle (42–182) vs 2L7B	Pocket 23: score 0.465, 884 Å^3, pLDDT 76.5, PAE 4.2 Å	**Possible but uncertain**

Table 1. *AlphaFold confidence, validation and pocket summary (from structure_assessment_ad.csv and the Lab 4 notebook).*

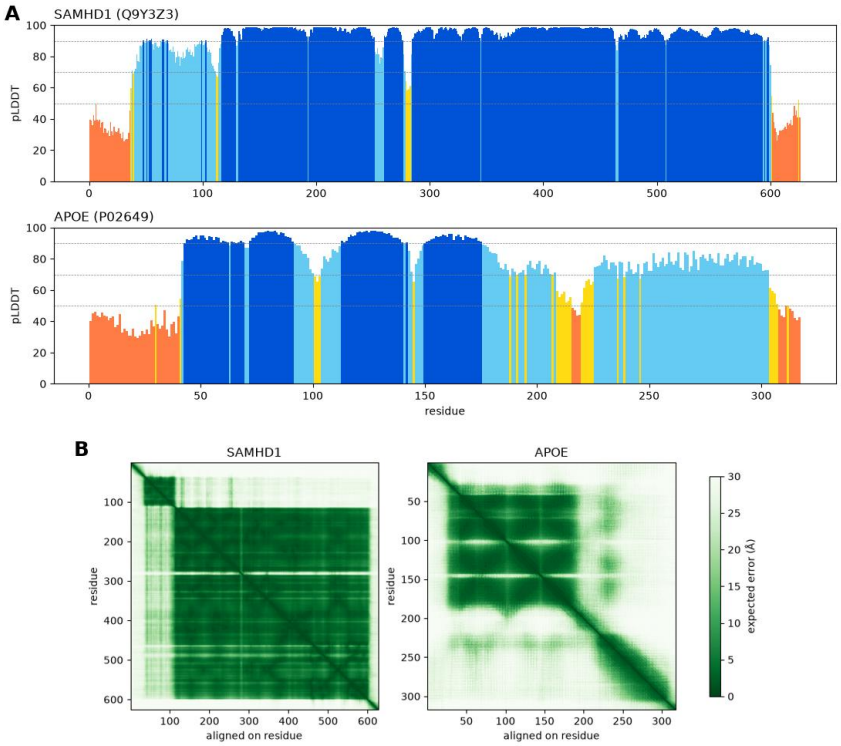


Figure 3. Predicted structure quality for the prioritised targets (AlphaFold DB models, Lab 4; Jumper et al., 2021; Varadi et al., 2022). *(A) Per-residue pLDDT (dark blue >90, light blue 70–90, yellow 50–70, orange <50). (B) Predicted aligned error; dark = confident relative position. SAMHD1 is one confident block; APOE shows a confident bundle with a pale, uncertain block between the bundle and the C-terminal domain.*

SAMHD1 has a high-confidence predicted structure that is also supported by an experimentally confirmed structure and a real binding pocket. The low-confidence regions shown in Figure 3A include the N-terminal tail (residues 1–36), the C-terminal tail (residues 602–624), and a short loop near residues 278–283. These regions are also missing or unresolved in the crystal structure, showing that the low confidence is consistent with their actual flexibility. The catalytic HD domain has a pLDDT score above 90 across almost the entire region, and the predicted structure matches the 4BZC crystal structure with an RMSD of 0.67 Å. The top pocket identified by fpocket (Le Guilloux et al., 2009) is located in this high-confidence region and covers all 17 residues that contact the catalytic-site dGTP analogue in 4BZC. Therefore, the pocket is supported by the model's confidence, an experimental structure, and a bound ligand. In comparison, the allosteric sites have scores of 0.003 or lower because, in a single-subunit model, each site forms only a shallow part of the pocket that normally exists at the interface between subunits (Ji et al., 2013).

The evidence for a drug-binding pocket in APOE is weaker, and the pLDDT scores help explain the reason. The four-helix bundle is predicted reliably and matches the 2L7B structure (Chen et al., 2011) with an RMSD of 2.60 Å. However, the predicted aligned error (PAE) between domains is 25.8 Å, as shown in Figure 3B. This indicates that the relative position of the C-terminal lipid-binding domain is uncertain. Therefore, the overall RMSD of 32.35 Å is not a meaningful measure of the model's accuracy. The selected pocket is formed by residues 31–56 and 163–174, near the receptor-binding region, but it only just meets the confidence requirements, with a mean pLDDT of 76.5. Part of the pocket wall consists of residues 31–40, which have pLDDT scores below 50 (Figure 4B). This low confidence means that these residues may not remain in the predicted position, so the actual pocket could be shallower or might not exist in the same form. Also, the 2L7B structure does not contain a bound ligand to confirm the

pocket. Therefore, I would not conclude that APOE has no drug-binding pocket, but I would also avoid designing drugs based on this predicted pocket until stronger structural evidence is available.

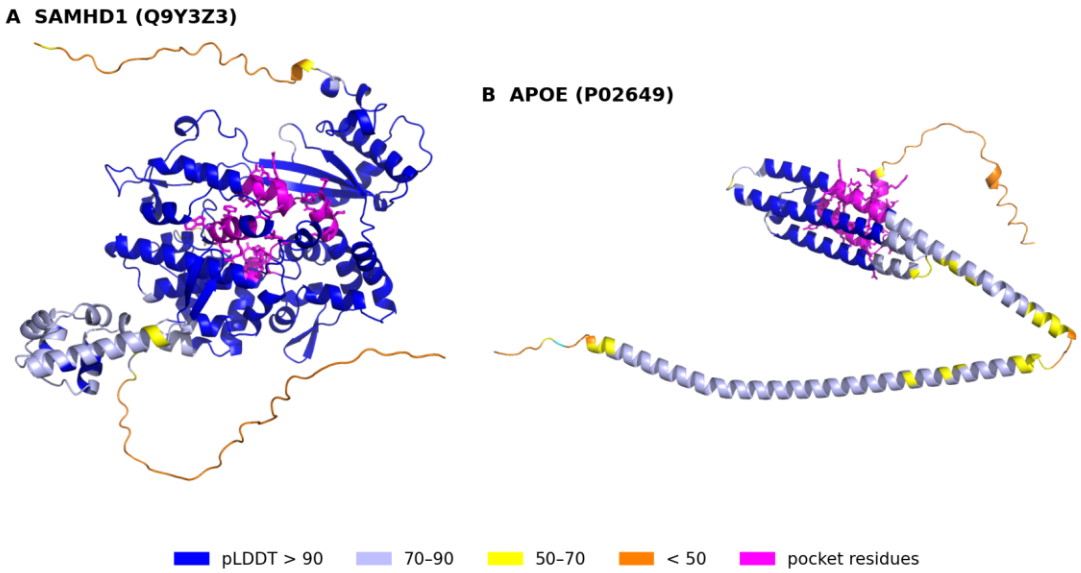


Figure 4. Predicted structure models of the prioritised targets *(AlphaFold DB, rendered in PyMOL), coloured by pLDDT; magenta sticks mark pocket residues. (A) SAMHD1: magenta residues are the catalytic-site dGTP contacts from 4BZC (149–150, 164, 207, 210, 215, 233–234, 311–319, 366–383), all lining fpocket pocket 3, buried in the confident (>90) HD domain. The lighter SAM domain (lower left) and orange tails are the low-confidence regions. (B) APOE: pocket 23 lining residues (31–56, 163–174) sit at one end of the confident four-helix bundle, next to the low-confidence N-terminus. The long, paler C-terminal helix is the domain whose placement is uncertain (inter-domain PAE 25.8 Å), so only the bundle supports a pocket claim. Pocket residues are shown in magenta regardless of their pLDDT.*

5. Prioritised target package

Table 2 is the list I would hand to a discovery team. The ranking rule is simple: a gene ranks higher when more labs independently point the same way, and a strong signal in one lab cannot make up for a missing or contradicting one.

#	Gene	Lab 2 evidence (RNA / protein / GWAS)	Lab 3 prior	Lab 4 structure	Modality and decision
1	**SAMHD1**	RNA −3.65 (adj p 1×10^{-4}), protein +0.05 (adj p 0.004), GWAS 2×10^{-36}; discordant; significant in all 3 layers	p_sm **0.824** (highest)	Pocket confirmed at catalytic site; 0.67 Å vs crystal	Small molecule. **Advance**, but settle direction of effect first (activator/stabiliser vs inhibitor)
2	**APOE**	Strongest AD GWAS signal ($p < 10^{-295}$); protein +0.15 (adj p 3×10^{-6}); RNA +1.17 (n.s.); concordant	Near zero (prior blind spot)	Bundle pocket, possible but uncertain	Antibody or level/lipidation modulator; small-molecule "structure corrector" is speculative. **Advance** as high-value, high-uncertainty
3	**CLU**	GWAS 1×10^{-47}; protein +0.14 (adj p 3×10^{-10}); RNA −0.92 (adj p 0.03); discordant; significant in all 3	p_ab 0.131 (secreted)	Not yet assessed	Antibody. **Next for structure**
4	**ASAP1**	Concordant down: RNA −1.48 (0.009), protein −0.08 (7.6×10^{-14}), GWAS 2×10^{-13}; significant in all 3	p_sm 0.193	Not yet assessed	Small molecule (weak prior). **Back-up**
5	**CACNG8**	Concordant down: protein −0.10 (3.4×10^{-9}), GWAS 2×10^{-10}; RNA −1.76 n.s. (0.078)	p_sm 0.380 + real clinical precedence	Not yet assessed	Small molecule; repurposing angle. **Back-up**

Table 2. *Final prioritised shortlist integrating all three layers. Fold changes are log2; p-values for RNA and protein are adjusted. Deprioritised from the top 15: APOC1 (GWAS signal largely from APOE linkage), MAPT (strong biology but already heavily pursued, disordered, near-zero priors), ZMYM2, PDZD2, WNK2, NUMA1, RHOJ, FARP2, COL11A1, NDUFA10 (weak protein significance, weak priors, or embeddings truncated at 2,046 residues).*

SAMHD1 is ranked first because all three labs support it. It ranks #1 in Lab 2 and is statistically significant in all three evidence layers, including a genome-wide significant GWAS association (Willett et al., 2025). It also has the highest small-molecule score in Lab 3 and a high-confidence catalytic pocket confirmed by a crystal structure in Lab 4. It is also one of the most interesting less-studied candidates. SAMHD1 is not a common AD-related gene; it is an innate immune enzyme that breaks down dNTPs. Loss of its function causes Aicardi–Goutières syndrome, an inflammatory brain disease driven by interferon (Rice et al., 2009). However, its AD results show a difference between RNA and protein levels. RNA expression is strongly reduced, while the protein shows a small change in the opposite direction. This difference may be due to the use of different patient groups and brain regions, or it may reflect a change specific to microglia that becomes less visible when measuring the whole tissue. This creates

an important concern: the confirmed binding pocket is the catalytic site, and blocking this site with an inhibitor could produce effects similar to SAMHD1 loss of function, which is linked to neuroinflammation. If reduced SAMHD1 activity contributes to AD, an activator or stabiliser may be a better treatment approach. Such a drug might target the allosteric dGTP-binding sites, which my single-subunit model could not score properly. Therefore, SAMHD1 remains the lead candidate, but the first question to answer is whether its activity should be increased or decreased in AD.

Why APOE Is Ranked Second Instead of First: APOE has the strongest genetic evidence among the candidates, and both its RNA and protein levels increase in the same direction. Therefore, based on its biological importance, APOE should be ranked first. However, it comes second because its druggability is still not proven. Both druggability scores are very low, and its predicted binding pocket depends partly on a low-confidence region. The structural analysis showed a possible pocket that the sequence-based model could not identify, but the evidence is still not strong enough to start drug-design work. CLU comes next because it is another AD-related gene with genetic support, but its RNA level decreases while its protein level increases. This may be related to its nature as a secreted and stable protein that accumulates in amyloid plaques. Its druggability score also shows a preference for antibodies. ASAP1 is the most consistent candidate across the evidence layers and is statistically significant in all three, but its druggability score is weak. CACNG8 has previous support for small-molecule drug development through TARP γ-8 modulators studied for epilepsy, although its RNA change is not statistically significant. I excluded APOC1 despite its high ranking because it is located close to APOE at chromosome region 19q13.32, and much of its GWAS signal may come from linkage with nearby genetic variants. I also excluded MAPT because, although tau is a well-established AD target, it has already received considerable research attention and the structural analysis did not identify a suitable binding pocket.

6. Limitations and experimental validation

There are some important limitations to this analysis. First, it cannot prove causality or show which direction of change would be useful for treatment. The three datasets come from different people and brain regions, so I can only say that a gene shows AD-related changes in independent studies, not that it directly causes the disease. Second, the bulk tissue data cannot identify the exact cell type responsible for the change. For example, it is not clear whether the change in SAMHD1 comes from microglia or neurons. Third, the Lab 2 scoring method uses the size of the effect without considering statistical significance. This can favour large RNA changes from the smallest dataset, which contains only 23 samples. Therefore, SAMHD1's first-place ranking depends partly on its RNA result. Fourth, the druggability scores are based on proteins that pharmaceutical companies have targeted before. The training dataset also contains around one-third positive examples, so these scores should be treated as rankings rather than actual probabilities of successful drug development. Fifth, the predicted structures represent single, static, wild-type proteins without bound ligands. They do not include the SAMHD1 tetramer, metal ions, or dGTP, and they do not represent lipidated APOE or the APOE ε4 variant. These limitations mean that the results need further experimental validation before making strong conclusions.

How I would validate SAMHD1: Step 1: Confirm the signal. I would first check SAMHD1 RNA and protein levels in samples from the same donors. For example, the ROSMAP dataset contains both bulk RNA-seq and TMT proteomics data. I would also use single-nucleus RNA-seq to identify which cell type is mainly responsible for the observed change.

Step 2: Determine the effect of SAMHD1. I would reduce SAMHD1 expression using CRISPR interference (CRISPRi) and increase its expression in induced pluripotent stem cell (iPSC)-derived microglia. I would then expose these cells to amyloid-β oligomers and measure interferon-stimulated genes, dNTP levels, cytokines, and phagocytosis. If reducing SAMHD1 increases inflammation, this would suggest that a drug activating SAMHD1 may be more useful than an inhibitor.

Step 3: Identify the correct binding pocket. I would model the SAMHD1 tetramer with bound dGTP and metal ions using AlphaFold-Multimer or Boltz-2. This would help assess the allosteric binding sites more accurately. Next, I would perform a fragment-based screening or thermal-shift assay using recombinant SAMHD1, along with a dNTPase activity assay, to identify potential compounds and determine whether they act as inhibitors or activators. For APOE, I would compare the ε4 and ε3 variants and run short molecular dynamics simulations to check whether pocket 23 remains stable when the flexible N-terminal region changes its position.

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